



Google Earth Engine (GEE)

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Google Earth Engine INFO



Register

Google Earth Engine

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Starting November 13, 2024, all Earth Engine access will [require a Cloud project](#). We will be limiting [quotas](#) for accounts without Cloud projects starting September 16, 2024.

A planetary-scale platform for Earth science data & analysis

Powered by Google's cloud infrastructure

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Meet Earth Engine

Google Earth Engine combines a multi-petabyte catalog of satellite imagery and geospatial datasets with planetary-scale analysis capabilities. Scientists, researchers, and developers use Earth Engine to detect changes, map trends, and quantify differences on the Earth's surface. Earth Engine is now available for commercial use, and remains free for academic and research use.

Create Project



Get started using Earth Engine

Earth Engine, Google's geospatial science platform in Google Cloud, is available for [paid commercial use](#) and [remains free for academic and research use](#). Learn more about [Google Cloud projects](#).

Let's get started:



Register a Noncommercial or Commercial Cloud project >

Have an existing project? [Click here to go to the Code Editor](#)

All users should register using a Cloud project. [Learn more](#).



How do you want to use Earth Engine?

☐ **Paid usage**

Commercial businesses, government operations. [See examples](#)

☒ **Unpaid usage**

Non-profits, education, government research, training, media.
[See examples](#)

Project type*

Academia & Research

Please note: If you will be accessing Earth Engine as a customer of a Google Cloud Platform reseller, please contact your reseller for terms and pricing governing your use of Earth Engine.

[BACK](#)

[NEXT](#)

Your information here is subject to [Google Cloud's Privacy Policy](#)



Create or choose a Cloud Project to register

Create a new project in Google Cloud, or choose one you are authorized to access to enable the API:

☒ **Create a new Google Cloud Project**

Organization

No organization

Project-ID*

ee-hosome2222

Choose a unique ID. This cannot be changed later.

Project Name (optional)

Earth Engine Default Project

Choose a name to help you identify the Cloud Project.

☐ **Choose an existing Google Cloud Project**

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CONTINUE TO SUMMARY

Landing Page

Google Earth Engine

Search places and datasets...

eee-hosome2222

Scripts Docs Assets

New Script

Get Link

Save

Run

Reset

Apps



Inspector

Console

Tasks

Filter scripts...

NEW



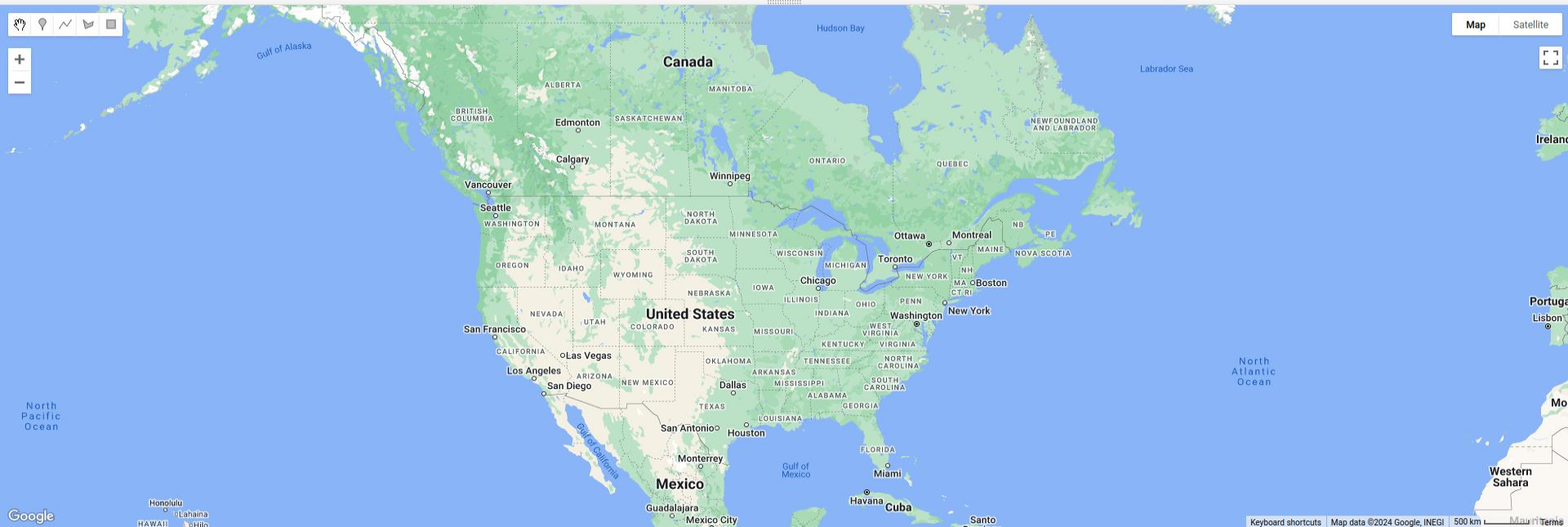
- Owner
- Writer
- Reader
- Archive
- Examples

1

Use print(...) to write to this console.

Welcome to Earth Engine!

Please use the help menu above (?) to learn more about how to use Earth Engine, or [visit our help page](#) for support.



Get Current Location Script

Google Earth Engine

Search places and datasets...

eee-hosome2222

Scripts Docs Assets

Filter scripts... NEW ↕

Owner

No accessible repositories. Click Refresh to check again.

Writer

No accessible repositories. Click Refresh to check again.

Reader

No accessible repositories. Click Refresh to check again.

Archive

No accessible repositories. Click Refresh to check again.

Examples

New Script *

Get Link Save Run Reset Apps ⚙

```
i 1 ui.util.getCurrentPosition(current_position, Error)
2
3 function current_position(point){
4   Map.addLayer(point,{color:'red'},"Current Position");
5   Map.centerObject(point);
6   print(point)
7 }
8
9 function Error(error){
i 10 print(error)
11 }
```

Inspector Console Tasks

Use print(...) to write to this console.

Point (29.99, 31.24) JSON

📍 📏 📐 📄 📁

+

-

Layers

Map

Satellite

El-Sayed El-Khouly

السيّد الخولي

Galal El-Sayed Haggag

جلال السيّد حجاج

Google

Keyboard shortcuts Map data ©2024 2 m

Terms

GEE Python API

package

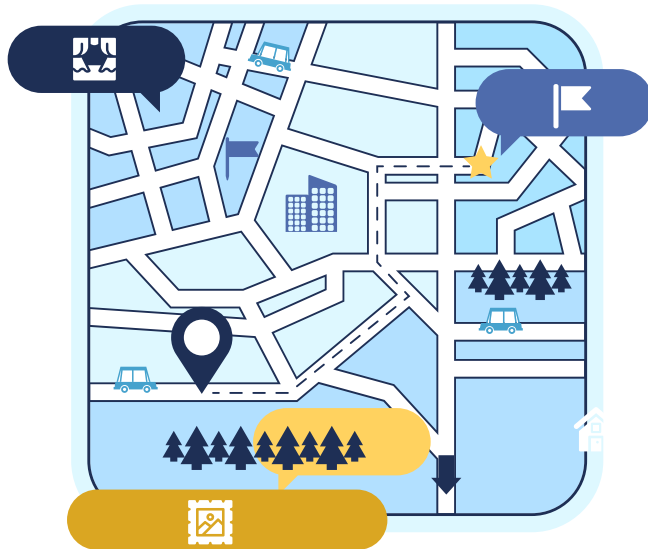
`pip install geemap`

`pip install earthengine-api`

libraries

`Import ee`

`import geemap`



Authentication

```
import ee
import geemap
```

[1] ✓ 0.9s

Python

```
ee.Authenticate()
```

[35] ⌚ 1m 51.5s

Python

...

...

To authorize access needed by Earth Engine, open the following URL in a web browser and follow the instructions:

https://code.earthengine.google.com/client-auth?scopes=https%3A//www.googleapis.com/auth/earthengine%20https%3A//www.googleapis.com/auth/cloud-platform%20https%3A//www.googleapis.com/auth/devstorage.full_control&request_id=HNdk0alSuawrs_suKtuyEtYSHhRoBRkEEQ77O3kj20c&tc=fclLK2YGyVPT3Hb17CEcfwnX7esHx2QY1i0DteYCW2I&cc=ImiqFl4_C1Dy4aO_2NFVEeSeS5ZBD

The authorization workflow will generate a code, which you should paste in the box below.

Authentication

Google Earth Engine

Notebook Authenticator

Active account: [redacted]@gmail.com

[SWITCH ACCOUNT](#)

Cloud Project: ee-hosome2727

[CHOOSE PROJECT](#)

Data access: ☐ Use read-only scopes

WARNING: ONLY PROCEED IF YOU NEED TO ACCESS EARTH ENGINE FROM A NOTEBOOK

The token that you generate here will allow access to your Google account. Ensure that you understand the notebook that you are running.

Any code that you include in the notebook (and anyone with access to the notebook kernel) will be able to copy or **change** your data. Enable read-only scopes above to prevent data changes.

If you are not running a notebook, or you don't understand these warnings, then the link that sent you here may be trying to trick you. Do not proceed!

GENERATE TOKEN

Granting permission. This creates a web application definition controlled by your project provided above. After you click Generate Token, Google will ask for your permission to grant the application access to your data. See details in the [step-by-step guide](#).

Expiry period. The granted permissions will expire in a week, after which you'll need to call `ee.Authenticate()` again.

Revoking permissions. You can view all applications connected to your account, and revoke permissions if needed, on <https://myaccount.google.com/permissions>. Search for "Earth Engine Notebook Client" to find the application defined by this page.

Technical details. The web application is defined by a development-mode "OAuth2 Client" on your specified project, which you can manage on the [Google Cloud Console](#).

Sign in with Google

Make sure you trust Earth Engine
Notebook Client -

[redacted]@gmail.com



Earth Engine Notebook Client -

[redacted]@gmail.com is requesting access to your Google Account. To protect your account, don't continue unless you know and trust this app.

Sign in or provide access to Earth Engine Notebook
Client - [redacted]@gmail.com

To sign in or provide access:

1. Copy the authorization code from the Authorization code section.
2. Navigate to Earth Engine Notebook Client - [redacted]@gmail.com.
3. Paste the authorization code on the Earth Engine Notebook Client [redacted]@gmail.com screen.

Authorization code

Please copy this code, switch to your application and paste it there:

4 / 1AVG

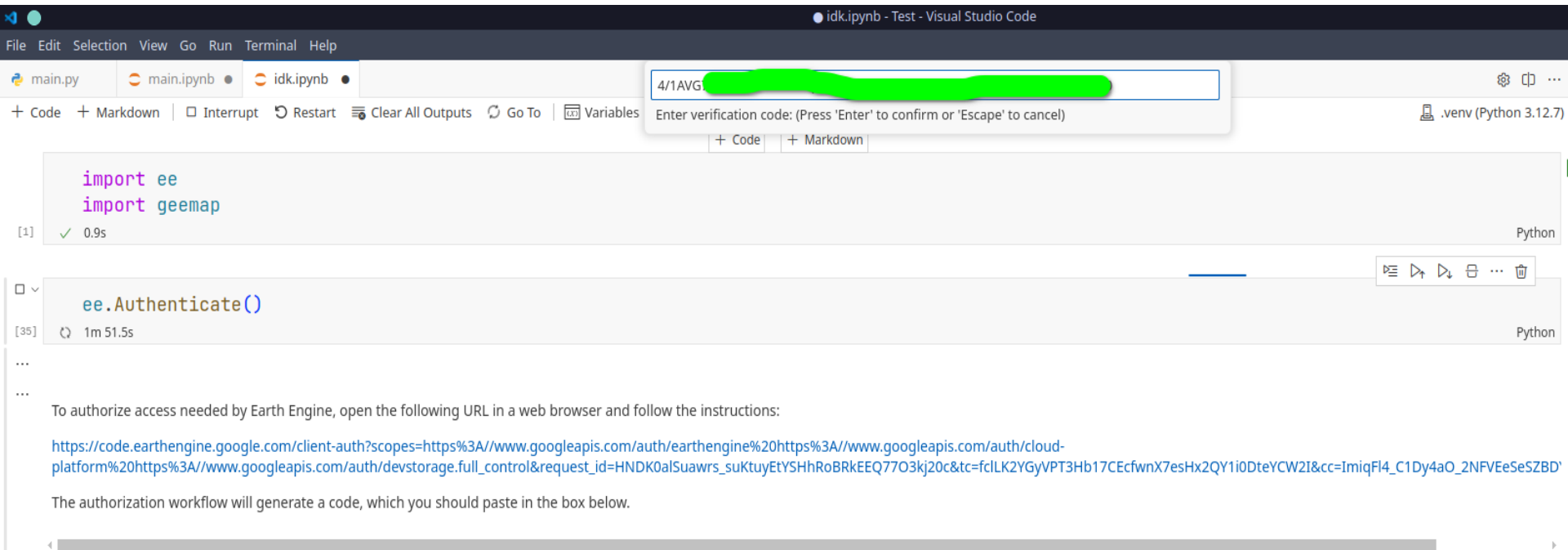
[redacted]



Don't sign in or provide access to Earth Engine
Notebook Client - [redacted]@gmail.com

If you don't want to continue, close this window.

Authentication



The screenshot shows the Visual Studio Code interface with a Jupyter Notebook open. The notebook has two cells. The first cell contains the code `import ee` and `import geemap`, which has been executed successfully, as indicated by a green checkmark and a 0.9s execution time. The second cell contains the code `ee.Authenticate()`, which has also been executed, with a 1m 51.5s execution time. A modal dialog box is displayed over the notebook, showing a verification code "4/1AVG" followed by a redacted area. The dialog prompts the user to "Enter verification code: (Press 'Enter' to confirm or 'Escape' to cancel)". The background of the notebook shows instructions for authenticating with Earth Engine, including a URL to open in a web browser and a note that the authorization workflow will generate a code to be pasted into the box below.

File Edit Selection View Go Run Terminal Help

main.py main.ipynb idk.ipynb

4/1AVG [REDACTED]

Enter verification code: (Press 'Enter' to confirm or 'Escape' to cancel)

+ Code + Markdown + Interrupt Restart Clear All Outputs Go To Variables .venv (Python 3.12.7)

```
import ee
import geemap
```

[1] ✓ 0.9s Python

```
ee.Authenticate()
```

[35] ⌚ 1m 51.5s Python

...

...

To authorize access needed by Earth Engine, open the following URL in a web browser and follow the instructions:

https://code.earthengine.google.com/client-auth?scopes=https%3A/www.googleapis.com/auth/earthengine%20https%3A/www.googleapis.com/auth/cloud-platform%20https%3A/www.googleapis.com/auth/devstorage.full_control&request_id=HNDK0alSuawrs_suKtuyEtYSHhRoBRKEEQ77O3kj20c&tc=fclLK2YGyVPT3Hb17CEcfwnX7esHx2QY1i0DteYCW2I&cc=ImiqFI4_C1Dy4aO_2NFVEeSeS5ZBD

The authorization workflow will generate a code, which you should paste in the box below.

Code for Testing

```
import ee
import geemap
```

[1] ✓ 0.9s

Python

```
ee.Authenticate()
```

[2] ✓ 0.4s

Python

True

```
ee.Initialize()
```

[3] ✓ 3.3s

Python

```
# Load an image.
image = ee.Image('LANDSAT/LC08/C02/T1_TOA/LC08_044034_20140318')

# Define the visualization parameters.
image_viz_params = {
    'bands': ['B5', 'B4', 'B3'],
    'min': 0,
    'max': 0.5,
    'gamma': [0.95, 1.1, 1],
}

# Define a map centered on San Francisco Bay.
map_l8 = geemap.Map(center=[37.5010, -122.1899], zoom=10)

# Add the image layer to the map and display it.
map_l8.add_layer(image, image_viz_params, 'false color composite')
display(map_l8)
```

[29] ✓ 1.6s

Python

Output



Change Cloud Project

Terminal

```
rm ~/.config/earthengine/credentials
```

GitHub Repository for Google Earth Engine Tutorials and Examples :

<https://github.com/giswqs/earthengine-py-notebooks/tree/master>

5. Examples

The Table of Contents below mimics the structure of the Google Earth Engine [API Documentation](#). I strongly encourage you to check out the API Documentation if you need an in-depth explanation of each Python example. Please note that the list below does not include all the Python examples contained in this repository. You are welcome to explore the repository and find more examples to suit your needs.

Get Started

- [Hello world!](#)
- [Adding data to Map](#)
- [Finding images](#)
- [Band math](#)
- [Mapping \(what to do instead of a for-loop\)](#)
- [Reducing](#)
- [Image statistics](#)
- [Masking](#)
- [A complete example](#)

```
Map = geemap.Map(center=[30.0444, 31.2357], zoom=8)
Map
gsw = ee.Image('JRC/GSW1_1/GlobalSurfaceWater')
occurrence = gsw.select('occurrence')
VIS_OCCURRENCE = {
    'min': 0,
    'max': 100,
    'palette': ['red', 'blue']
}
VIS_WATER_MASK = {
    'palette': ['white', 'black']
}
water_mask = occurrence.gt(90).selfMask()
Map.setCenter(31.2357, 30.0444, 8)
Map.addLayer(occurrence.updateMask(occurrence.divide(100)), VIS_OCCURRENCE, "Water Occurrence (1984-2018)")
Map.addLayer(water_mask, VIS_WATER_MASK, '90% occurrence water mask', False)
color_bar_params = {
    'title': 'Water Occurrence (%)',
    'min': 0,
    'max': 100,
    'palette': ['red', 'blue']
}
Map.add_colorbar(color_bar_params)
Map.addLayerControl()
Map
```



THANK YOU !